RAFIUL RABBI ZAHID

Artificial Intelligence Enthusiast | ML Researcher

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Professional Summary

Final-semester IoT and Robotics Engineering undergraduate with hands-on experience building and evaluating machine learning systems across IoT, Robotics, healthcare NLP, computer vision, and embedded AI. Author of four peer-reviewed and accepted publications at IEEE and Elsevier-indexed venues, covering Retrieval-Augmented Generation, transformer-based classification, and ensemble ML frameworks. Proficient in Python and PyTorch, with practical experience in transfer learning, CNN architectures, and LLM/RAG integration. Quick to adopt new tools and frameworks, with a track record of translating research prototypes into working, evaluated systems. Seeking a Intern/Trainee Engineer role to apply research-driven problem-solving to production ML and software engineering projects.

Technical Skills

Programming Languages: Python, C, C++

Machine Learning & Deep Learning: PyTorch, Transfer Learning, CNN Architecture Design, Multi-Label Classification, Ensemble Learning

NLP & LLM: Transfer Learning, Retrieval-Augmented Generation (RAG), LLM Integration, Vector Search

Data & Analytics: NumPy, Pandas, Matplotlib, Data Preprocessing, Exploratory Data Analysis

Databases: MySQL, SQLite, FAISS, Chroma (Vector Databases)

Embedded & IoT Systems: Arduino, ESP32, Sensor Integration, Actuator Control, Real-Time Data Acquisition

AI & Automation Tools: ChatGPT, Claude, n8n Workflow Automation, Zapier, Make

Platforms & Tools: Linux, Windows, LaTeX, Git-based collaboration, Microsoft Office

Experience

Research Assistant — Elite Research Lab LLC *Dhaka, Bangladesh | May 2026 – Present*

- Conduct applied research across medical AI, computer vision, NLP, and edge AI, producing publication-ready experimental results for peer-reviewed venues.
- Design and run experiments on constrained compute (Google Colab, T4 GPU), optimizing model architectures and training pipelines for efficiency.
- Collaborate with faculty and student co-authors to prepare manuscripts, respond to reviewer feedback, and finalize submissions for IEEE and Elsevier journals/conferences.

Key Projects

Multilingual Toxic Content Detection System *Python, mBERT, XLM-RoBERTa, Flask | Feb 2026*

- Engineered a multilingual toxicity-detection system for Bangla and Banglish text using transformer-based transfer learning.
- Built a multi-label classification pipeline to identify multiple toxicity categories within a single message.
- Deployed a Flask-based inference API enabling real-time content moderation.

IoT & ML-Enabled Smart Vertical Farming System *ESP32, Machine Learning, Sensors | Jul 2025*

- Developed an ESP32-based IoT system for real-time monitoring of vertical farming environmental conditions.
- Built a machine learning model to recommend suitable crops from sensor data, improving resource and water-use efficiency.
- Findings published as a co-authored paper at IEEE QPAIN 2026.

RAG Healthcare Assistant *RAG, Machine Learning, Hugging Face, NLP | Jan 2025*

- Designed a Retrieval-Augmented Generation (RAG) healthcare assistant to deliver evidence-based responses to patient queries.
- Integrated an ML-based disease-prediction module with symptom analysis to strengthen the assistant's clinical relevance.

Intelligent Email Response Automation System *n8n, Gmail API, OpenAI, Workflow Automation | May 2026*

- Built an n8n workflow that classifies incoming Gmail messages and generates AI-drafted replies via OpenAI integration.
- Designed category-specific reply logic to automate inbox triage and reduce manual response time.

Publications

Conference Papers

- Rakibul Hasan Rakib, **Rafiul Rabbi Zahid**, Suman Saha, "VertiCulture: An Integrated IoT and Machine Learning Enabled Smart System for Optimizing Vertical Farming Environments." In *Proceedings of the 2026 IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence & Networking (QPAIN)*, IEEE, 2026.
- **Rafiul Rabbi Zahid**, Abidul Islam Alif, Shahbaj Ahmed Sakib, Tajbir Ahmed, Abdul Hasib, Mostafiz Ahammed, "Environmental Drivers of Lumpy Skin Disease: An Interpretable Ensemble Machine Learning Framework for Climatic Risk Assessment." In *Proceedings of the International Conference on Electrical, Computer and Communication Technologies (ECCT 2026)*, 2026. **(Accepted)**
- **Rafiul Rabbi Zahid**, Abidul Islam Alif, Shahbaj Ahmed Sakib, Maria Akter Mim, Md. Abu Horaira Ansary Tanvir, Abdul Hasib, "An Explainable and Uncertainty-Aware Machine Learning Framework for Automated Smoking Risk Assessment Using Clinical Health Indicators." *International Conference on Power, Electronics, Communications, Computing, and Intelligent Infrastructure (PECCII 2026)*, Pabna University of Science and Technology (PUST), 2026. **(Accepted)**

Journal Articles

- **Rafiul R. Zahid**, Rakibul H. Rakib, Iftekhar Rahman, Md. A. Amin, Tanjum R. Akanto, Fahmida A. Antara, Sadia Enam, "IoT for Industrial Automation: A Review on Industry 4.0 Applications." *International Journal of Information Engineering and Electronic Business (IJIEEB)*, MECS Press, 2026. DOI: 10.5815/ijieeb.2026.02.11

Education

B.Sc. (Engg.) in Internet of Things and Robotics Engineering *Apr 2022 – Present*

University of Frontier Technology, Bangladesh (UFTB), Gazipur | CGPA: 3.47 / 4.00

Higher Secondary Certificate *2018 – 2020*

Government Edward College, Pabna | GPA: 5.00 / 5.00

Achievements & Awards

- Dean's List Award, Faculty of Cyber Physical Systems Engineering, UFTB — Outstanding Academic Performance, Level 3 (Jul 2026).
- 1st Runner-Up, IoTrix — Televerse 1.0 National-Level Hackathon, organized by Dept. of ETE, CUET (Nov 2025).
- 2nd Runner-Up, Espectrum 1.0 Project Showcasing Event, organized by RUET IoT Club (Apr 2025).

Leadership & Professional Involvement

- Executive Member, Basis Student Forum – UFTB Chapter (Mar 2024 – Jun 2025).
- Executive Member, UFTB Debating Society (May 2023 – Aug 2024).
- Student Activity Member, UFTB Cloud Club (Nov 2023 – Aug 2024).

Certifications

- 21st Century Employability Skilling Program — Wadhwani Foundation (Feb 2024): critical thinking, problem-solving, communication, and teamwork.

References

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